

GENIE PRO STUDIOS

Product Requirement Document

A full specification of features, architecture, booking logic, site structure, and commercial terms for the Genie Pro Studios web application.

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PREPARED FOR

Genie Pro Studios

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STATUS

01

Business Overview

Genie Pro Studios is a music studio offering bookable recording, rehearsal, and production sessions. The web application exists to remove friction from the booking process: customers should be able to see everything they need to know, book a session, agree to terms, and pay — without back-and-forth conversations with the studio owner for routine bookings.

- Business type: Music studio (recording / rehearsal / production sessions).
- Primary goal: a self-service landing experience that front-loads the information customers usually ask about, so the owner isn't repeating the same answers.
- Secondary goal: legally sound consent — customers must explicitly agree to Terms & Conditions before payment, with a timestamped record, to prevent "I didn't see it" disputes.
- Owner is notified automatically whenever a new booking is made.

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Core Requirements

Must-have features

- Landing page that condenses all essential information (services, pricing, policies) so customers rarely need to ask directly.
- Terms & Conditions consent flow — required, explicit agreement before payment, with a timestamped proof-of-acceptance record.
- Appointment / session booking system with date & time slot selection.
- Owner-controlled availability — the studio owner manually opens specific dates and time slots; customers can only select from what has been explicitly opened.
- Automatic slot locking — once a slot is booked, it is immediately removed from the available pool to prevent double-booking.
- Owner notification on every new booking (channel: email / SMS / WhatsApp — final channel to be confirmed).
- Online payment via Paystack, confirmed server-side before a booking is finalized.

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Technical Architecture & Stack

Frontend

- Next.js + TypeScript — application framework and type safety across the app.
- Tailwind CSS — utility-first styling.
- Shadcn/ui — used selectively where needed (forms, dialogs, calendar/date-picker), not wholesale.
- Framer Motion — page and interaction animations/transitions.
- React Hook Form + Zod — form state management and schema validation; the same schema can be reused to validate on the server, keeping client and server in agreement on what "valid" means.

Data-access pattern

A layered pipeline is used on the frontend so each layer only knows about the one directly below it:

Data flow

```
Component  -> Hook  -> Service  -> Repository  -> Supabase / Node API
(UI only)   (state) (business (raw data (actual database
              logic)  access)    / API)
```

- Repositories (classes) — the sole place that talks to Supabase/the API. Dumb data access only, e.g. BookingRepository, AvailabilityRepository.
- Services — business logic and orchestration above repositories, e.g. BookingService.book() checks slot availability, confirms T&Cs acceptance is recorded, creates the booking, and triggers the owner notification.
- Hooks — React-facing glue (useBooking(), useAvailability()) that call services and expose state/loading/errors to components, keeping components purely presentational.

Rationale: swapping backend technology, or changing a business rule, touches exactly one layer — never the UI.

Backend

- Supabase — Postgres database, Auth (customer/owner accounts), Row Level Security (safe direct client queries), and Storage if file uploads are needed.
- Node.js + TypeScript — a thin custom backend for what Supabase should not handle directly from the client: Paystack integration and webhook handling, owner notifications, and recording proof of T&Cs acceptance.

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Booking & Availability Logic

Availability management

- The owner manages availability fully manually, date-by-date — opening a calendar view and marking specific dates and time slots as available, rather than relying on a recurring weekly template. This trades a little extra admin effort for full control.
- Customers only ever see and can select dates/slots the owner has explicitly opened; everything else is greyed out or hidden in the date picker.
- Once a slot is booked, it is immediately removed from the available pool — no double-booking is possible.

End-to-end booking flow

Booking flow

```
Select service
-> Select date (only owner-opened dates selectable)
-> Select time slot (only open slots shown)
-> Enter customer details (React Hook Form + Zod)
-> Review condensed T&Cs + consent checkbox (required)
-> Pay via Paystack
-> Backend verifies payment via Paystack webhook
-> Booking confirmed + owner notified
```

Booking is only marked confirmed after the Paystack webhook verifies successful payment server-side — never on the client-side redirect alone, since a redirect can be spoofed, dropped, or interrupted before the money has actually cleared.

05

Site Map — Pages & Structure

The map below assumes guest booking (customers provide name, email and phone at booking time; no account or login is required) to keep the booking flow as frictionless as possible, consistent with the goal of reducing what customers need to ask the owner directly. If account-based booking is preferred instead, this section would gain Sign Up / Login / Customer Dashboard pages.

Public-facing pages

Page	Route	Purpose
Home / Landing	/	Hero, condensed essentials (services, pricing, key policies), gallery/testimonial preview, primary CTA into the booking flow.
Services	/services	Bookable session types, pricing per type, what's included (engineer, equipment, room). Links into booking pre-filtered to a service.
About	/about	Studio story, the space, the team/engineer(s) — builds trust before payment.
Gallery / Portfolio	/gallery	Photos/videos of the studio and past work.
Book a Session	/book	The core page — multi-step flow: service, date, time slot, customer details, condensed T&Cs + consent, Paystack payment.
Booking Confirmation	/book/confirmation	Shown after payment is verified server-side. Booking summary, reference number, calendar add-on, studio contact info. Triggers owner notification.
Find My Booking	/booking/lookup	Guest-friendly lookup (email/phone + reference number) to check a booking without needing an account.

Page	Route	Purpose
Terms & Conditions	/terms	Full legal text — source of truth the condensed in-flow version links back to.
Privacy Policy	/privacy	Standard companion to Terms, required given personal data and payment processing.
Contact	/contact	Location/map, contact form, socials, studio hours for pre-booking questions.
404	—	Catch-all with a path back to Home / Book a Session.

Admin-facing pages (owner only, Supabase Auth)

Page	Route	Purpose
Admin Login	/admin/login	Owner-only sign-in. Not linked from public navigation.
Availability Manager	/admin/availability	Owner's calendar to manually open/close specific dates and time slots — powers what's selectable on the public booking page.
Bookings List	/admin/bookings	All bookings, filterable by date/status, with payment status at a glance.
Booking Detail	/admin/bookings/[id]	Full detail on one booking: customer info, service, payment status/reference, and the timestamped T&Cs acceptance record.

How the pages connect

Site flow

```
Home --+--> Services --+
      +--> About      |
      +--> Gallery    +--> Book a Session --> Confirmation --> (owner notified)
      +--> Contact    |
      +--> Terms/Priv--+ +--> Booking Lookup (anyone with the reference)

Admin Login --> Availability Manager --(controls what's selectable in)--> Book a
Session
                        |
                        +--> Bookings List --> Booking Detail
```

The core loop: the owner opens availability in the Availability Manager, which determines what customers can select on the Book a Session page. A completed, paid, and consented booking appears in the Bookings List and notifies the owner, while the customer can always re-check it via Booking Lookup. Every public page ultimately funnels toward Book a Session — the one page doing the real work — while Home, Services, About, Gallery, and Contact exist to build enough trust and answer enough questions upfront that the customer doesn't need to message the owner directly before booking.

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Payment & Commercials

Payment integration

- Payment provider: Paystack.
- The Node.js backend handles the Paystack webhook (payment confirmation) server-side.
- A booking is only marked confirmed after the webhook verifies successful payment — never on the client-side redirect alone — so an interrupted or spoofed redirect can never falsely confirm a session.

Project cost

Item	Amount
Development fee	NGN 250,000
Setup, environment & production fee (includes domain, database, and post-production setup)	NGN 70,000
Total	NGN 320,000

These figures are internal commercial terms between the developer and Genie Pro Studios — not customer-facing site content.

Payment schedule

Before commencement:	70% of total project cost (NGN 224,000)
After completion:	30% of total project cost (NGN 96,000)

Payment account details

Account number:	9133328567
Bank:	Opay
Account name:	Omotosho Peter Oluwadarasimi

Post-handover maintenance

Maintenance clause

Following project handover, the developer will provide maintenance support (bug fixes and minor adjustments to delivered functionality) for a period of 2 months at no additional cost.

Any work beyond this window, or requests for new features/scope not covered in this document, will be scoped and billed separately.

07

Open Items & Assumptions

To be confirmed before / during build

Final list of bookable session types (recording, mixing/mastering, rehearsal, vocal booth, etc.) and whether pricing is flat, hourly, or per engineer/room.

Whether a sound engineer is assigned per booking, or it is purely room/equipment rental.

Guest booking (assumed in this document) vs. account-based booking with customer login.

Notification channel for the owner: email, SMS, WhatsApp, or in-app — does not change the site map, but affects what fires on booking confirmation.

Branding: visual identity, color palette, and typography have not yet been defined.

